

Design and Development of Novel Haptic Skin Strain Device for the Arm

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ABSTRACT

Skin strain is an understudied haptic stimuli that has great potential due to its similarity to touch sensations that are commonly felt in daily life. However, up until now, applications have been focused more on smaller areas of the body, and have focused more on skin strain at an actuator level rather than a garment level. As haptics continues to grow as a field, understanding how stimuli are perceived at a micro and macro scale will be essential to producing innovative haptic products that meet a variety of user needs. This work begins an exploration of artificially produced skin strain on a larger area of the body, covering early development work on a skin strain system for the full arm. The motivation and design details are discussed, and then early work related to actuator optimization and plans for future research.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices; • **General and reference** → Cross-computing tools and techniques; Design.

KEYWORDS

Wearable Technology, Haptics, Skin Strain, Shape Memory Actuators

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1 INTRODUCTION

The use of haptic stimuli has been a prominent area of focus for the development of wearable technologies, especially for soft robotics that combine novel technologies with traditional apparel manufacturing techniques. An understudied method of haptic stimuli (relative to modes like vibration and compression) is skin strain, where specific areas of the skin are displaced. This work focuses on the initial development of a novel artificial skin strain system for the full arm, leveraging SMA as the primary form of actuation. The goal is to understand artificial skin strain as a full-fledged system rather

than exploratory use cases, with a focus on engineering tradeoffs, system design factors, and user experience, including: garment materials, actuator integration, perception differences around the arm, and the body-product relationship. In this sense, this project will provide design information at the garment level in addition to the actuator level.

The literature on skin strain demonstrated clear potential for skin strain for wearable devices while also demonstrating areas for future exploration and growth. For example, Hamden et. al. [2] developed an adhesive SMA actuated skin strain system that was able to present a multitude of unique skin strain sensations. The key issue is the reliance on adhering to this skin, which can cause discomfort and is therefore not easily scaled to larger areas of potential actuation. As well, Muthukumarana et.al. [1] showed potential for skin strain on a larger area of the body, and how these larger actuator formations may work in practice. In addition they showed that silicone provided comparable results to adhesion as an anchoring method for skin strain, however, this was limited to plain wire SMA in a sinusoidal formation. Clark, Kim, & O'Malley [6] Found that skin strain was perceived differently depending on the direction of the stimulus in their servo motor based skin strain system. The directions were vertically along the length of the arm and horizontally around the circumference of the arm. The drawback is the limitation of device placement on the body, where they chose to keep the garment at one area of the arm. As well, diagonal directions were not included in their study.

These advancements in skin strain technology demonstrate the potential for haptic systems that utilize some form of skin strain. However, there are areas for further advancement, including: an thorough examination of the perception of SMA generated skin strain along different directions, further study of how increasing the potential area of actuation works on the body, and further validating alternatives to adhesion as a skin strain anchoring method. As the haptics market moves towards more garment integrated experiences, these key insights will be essential to producing quality products.

2 DEVICE DEVELOPMENT

In order to produce the desired haptic effect several key considerations must be accounted for. First, attachment of the actuators must be close to the skin and strong enough to perform the desired displacement. A garment that is too loose could be construed as a brushing sensation and inversely a tight attachment may feel more like compression. As well, linear actuators typically need two points of contact in order to create the desired strain. In addressing these needs the garment shown in Figure 1 was developed

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Figure 1: (A) Prototype on the Arm (B) Close-Up Photo of Actuators

The garment is currently made for the forearm, expanding to the full arm later, and is being evaluated for optimal sizing of the actuator components and refining the user experience of the garment. A crepe scuba knit fabric is attached to a 2mm thick silicone layer (Smooth-On Mold Max 10T) and then cut to the garment pattern shape. Next, shape memory alloy (SMA) spring actuators – which contract linearly when heated, creating tension/strain in the conjoined fabric – are then attached to the garment. This combination of materials provides the necessary thermal insulation to protect the wearer from heat generated by the SMA actuators, while remaining thin enough to better interface with the skin. Importantly, it moves skin strain away from adhesive applications, which are difficult to don and doff, and cause discomfort. As mentioned, Muthukumarana et.al. [1] found silicone to be a viable alternative to skin adhesion. Therefore the present work seeks to reaffirm this finding and expand our understanding of silicone paneling in a larger system. It will also see how SMA coil actuators perform with silicone anchoring.

The actuators utilize SMA in a heat-set, coiled formation (Dynalloy Flexinol @70°C (0.012" diameter wire)) which were then cut to one of 3 different lengths (1in, 2in, or 3in). SMA coils were chosen for their low-profile on the body and high force output [4], and their use in previous skin strain studies [1, 3]. The ends are then crimped into a metal zipper stop with a hole that allows it be attached to the hook part of a metal hook-and-eye attachment that is sewn to the fabric. This novelty of this method extends on methods used in previous work. Systems like the one developed by Hamdan et.al. [2] have utilized crimping to attach SMA directly to wiring, however, the added use of a hook-and-eye attachment here allows for the SMA to be removed for replacement/repair without disturbing the rest of the system. Other common methods, like snaps [5], have also been shown to be useful for holding SMA but tend to lift up the ends above the fabric. This lifting is undesirable for skin strain applications since the actuator needs a close relationship with the skin to properly displace it and produce skin strain.

3 EARLY TESTING PLAN AND CONCLUSION

Early perceptive testing has focused on comparing the actuator lengths across the arm to determine which is most consistently identified as skin strain. Preliminary testing indicates that the optimal length of SMA actuators for generating perceptible skin strain stimuli is between 1" and 2" condition, with both being readily

perceived as skin strain. The 3" condition is easy to mistake for a compression stimuli because of how wide an area the stimuli actuates across.

Future development will focus on expanding the design to cover the full arm, and conducting user testing to assess research questions related to how the perceptual experience of skin strain on the arm changes between different actuator formations and locations. This includes placing actuators at different angles to check for perceptive differences, and understanding how perception changes at different regions along the arm at these different angles. As mentioned earlier, previous work has indicated directional effects for the perception of skin strain [6]. However, this was for a servo motor based system and was limited to just horizontal and vertical directions in one arm location. Activation timing and speed need to be refined as well to ensure the stimuli is fast enough to be readily perceived as skin strain. Overall, the expected contribution of this device will be to conceptualize skin strain at the garment level in addition to the actuator level. Allowing for a refined understanding of wearable skin strain in action.

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